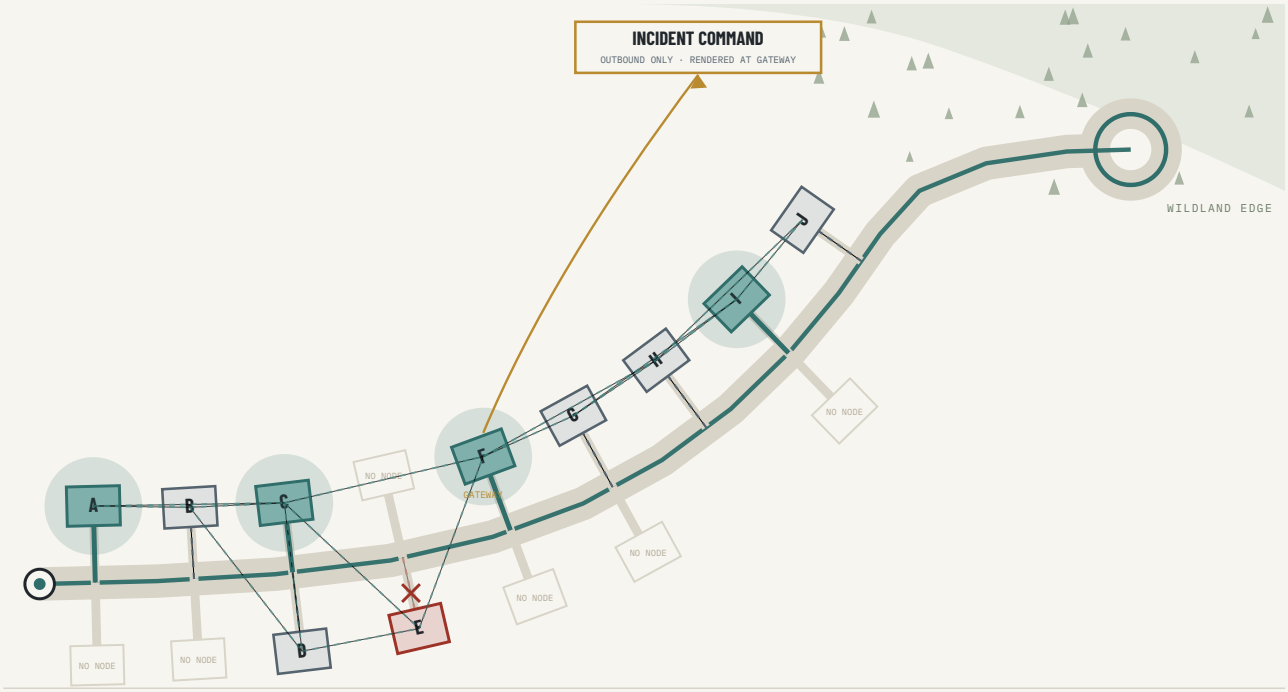
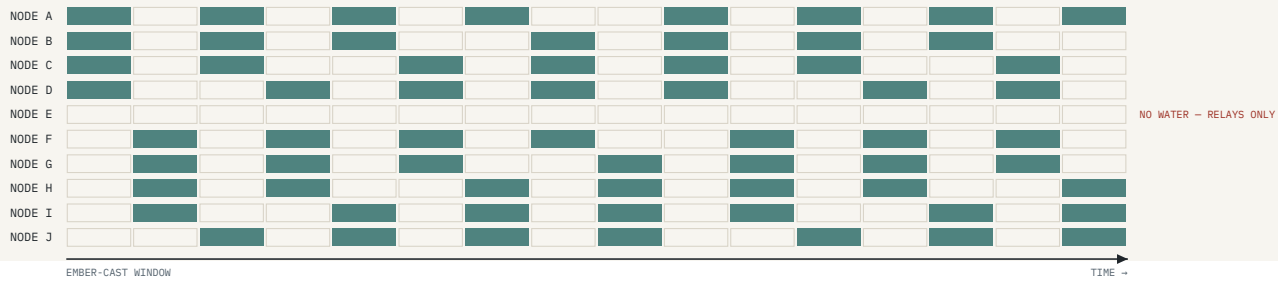
FIG. 1 — THE COORDINATION DOMAIN

A NEIGHBORHOOD SHARING ONE SUPPLY · PLAN VIEW · PARTIAL ADOPTION



THE ROTATING SCHEDULE

EACH NODE DECIDES LOCALLY FROM SHARED STATE · NO COORDINATOR · NO ELECTION · FOUR MAY DRAW AT ONCE



SUPPLY SAFETY
AGGREGATE DRAW NEVER EXCEEDS SUSTAINABLE CAPACITY

COVERAGE FLOOR
NO STRUCTURE EXCEEDS ITS MAXIMUM DRY INTERVAL

PARTIAL ADOPTION
UNEQUIPPED STRUCTURES DO NOT STOP THE DOMAIN WORKING

FIG. 2 — THE LAYER MODEL AND THE TRUST BOUNDARY

COORDINATION NEVER NAMES A RADIO · NOTHING LEAVES EXCEPT THROUGH PUBLICATION POLICY

DEPENDENCIES POINT DOWNWARD ONLY

OSI 7 — coord/ state/ ui/ ledger/ app/

APPLICATION

Coordination · state machine · degradation · operator UI · ledger

OSI 5-6 — proto/ policy/

VOCABULARY AND TRUST

Closed verb enumeration · signing · publication policy

SUBSTITUTION LINE — SWAP ANYTHING BELOW

LORA · MANET · WIRE · WHATEVER COMES NEXT. NOTHING ABOVE CHANGES.

OSI 4 — proto/

ENVELOPE

Compact binary envelope · best-effort delivery

OSI 3 — link/

NETWORK

Mesh addressing and relay

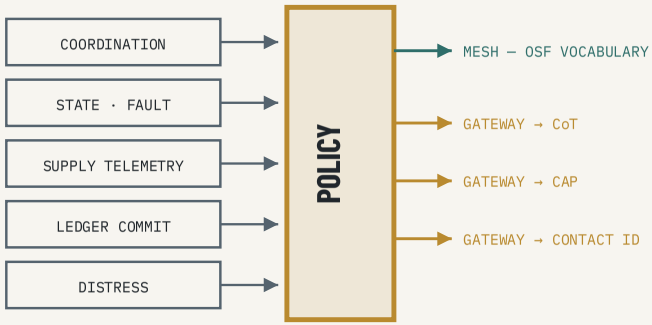
OSI 1-2 — hal/ link/

PHYSICAL AND LINK

Radio hardware · framing · the wire itself

THE CHOKEPOINT

EVERY OUTBOUND BYTE PASSES ONE MODULE



INBOUND

VERIFIED BEFORE IT IS ANYTHING ELSE

1. SIGNATURE VERIFIED
2. SCOPE RESOLVED LOCALLY
3. EXPIRY HONOURED — AUTO-RESUME
4. CLOSED VERB, OR DROPPED

NO INBOUND MESSAGE MAY DISABLE AN OPERATOR CONTROL.

TRANSPORT INDEPENDENCE

COORDINATION IMPORTS AN INTERFACE, NEVER A RADIO.
A LINK CHANGE CANNOT REACH LAYER 7.

PUBLICATION POLICY

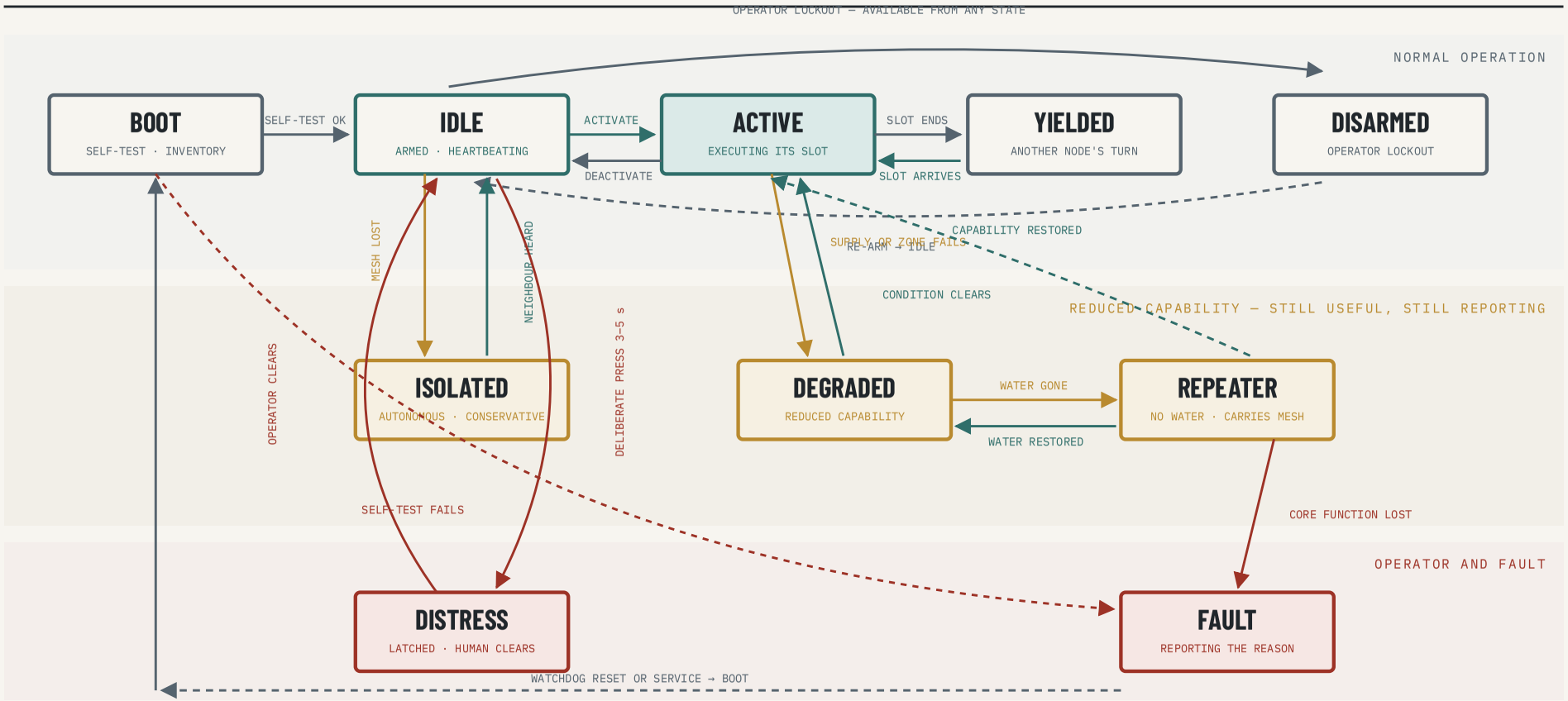
THE TRUST LAYER SITS ABOVE TRANSPORT BECAUSE IT IS
A PROPERTY OF THE BOUNDARY, NOT OF A FEATURE.

NO SESSION STATE

BEST-EFFORT BY DESIGN. A MISSED MESSAGE IS NORMAL;
SUPPLY PRESSURE IS THE FALLBACK.

FIG. 3 — THE NODE STATE MACHINE

EVERY FAILURE HAS A DEFINED DESTINATION · EVERY DESTINATION HAS A WAY BACK



POWER LOSS IS NOT A STATE

SPRING-RETURN VALVES CLOSE BY PHYSICS. SOFTWARE DOES NOT PARTICIPATE AND MUST NOT ASSUME IT CAN.

ISOLATED IS NOT AN ERROR

MOST NODES SPEND MOST OF THEIR LIVES WITH INTERMITTENT CONTACT. A NODE THAT PANICS ALONE IS BROKEN.

REPEATER EARNS ITS PLACE

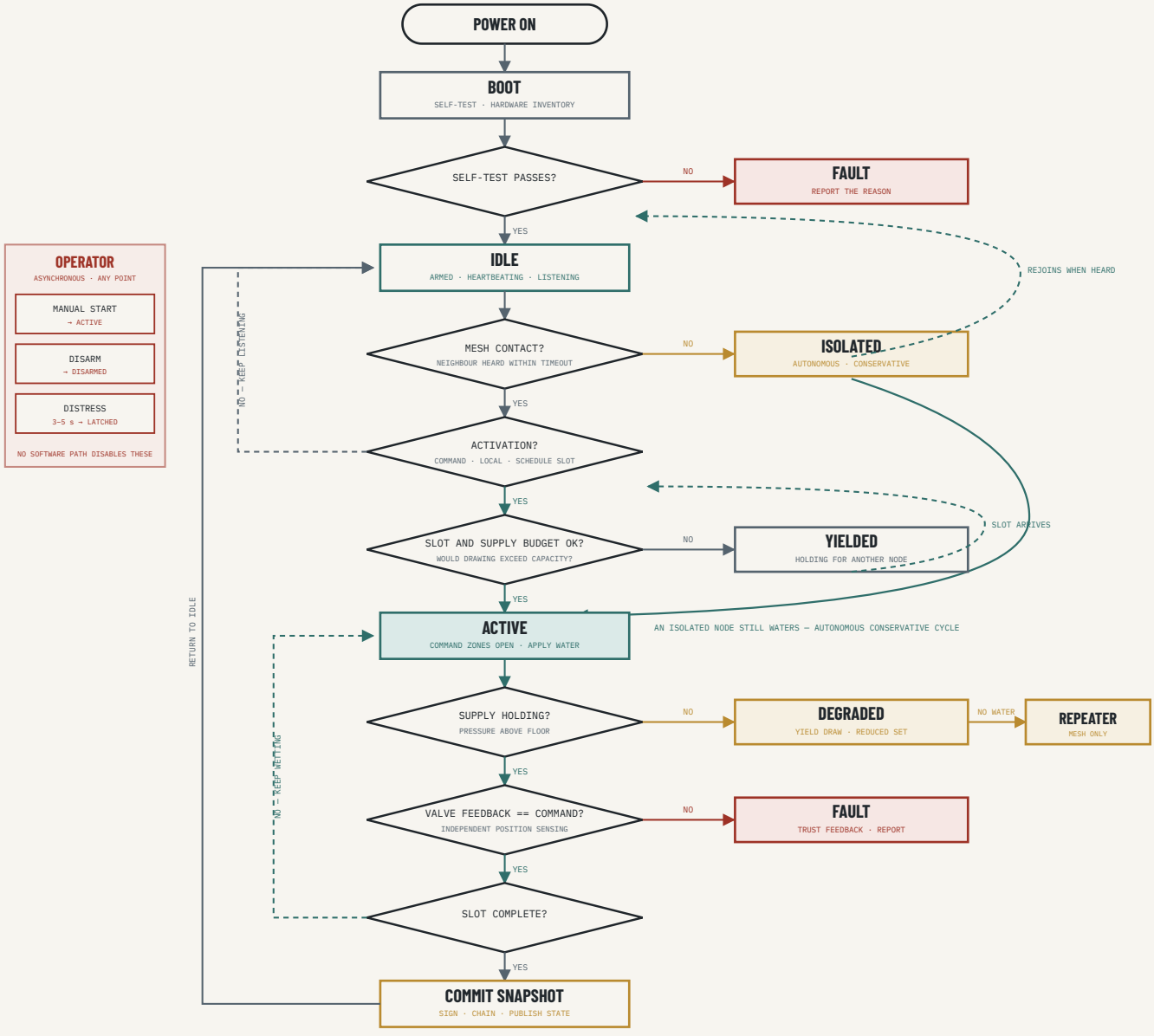
A NODE WITH NO WATER STILL CARRIES ITS NEIGHBOURS' TRAFFIC. IT IS WORTH LEAVING ON THE WALL.

NOTHING IS A DEAD END

EVERY DEGRADED STATE HAS A RETURN PATH. DEGRADATION IS A ROUTE, NOT A TERMINUS.

FIG. 4 — NODE OPERATION, CONTROL FLOW

THE SAME STATE MACHINE READ AS SEQUENCE · COMPANION TO THE STATE DIAGRAM

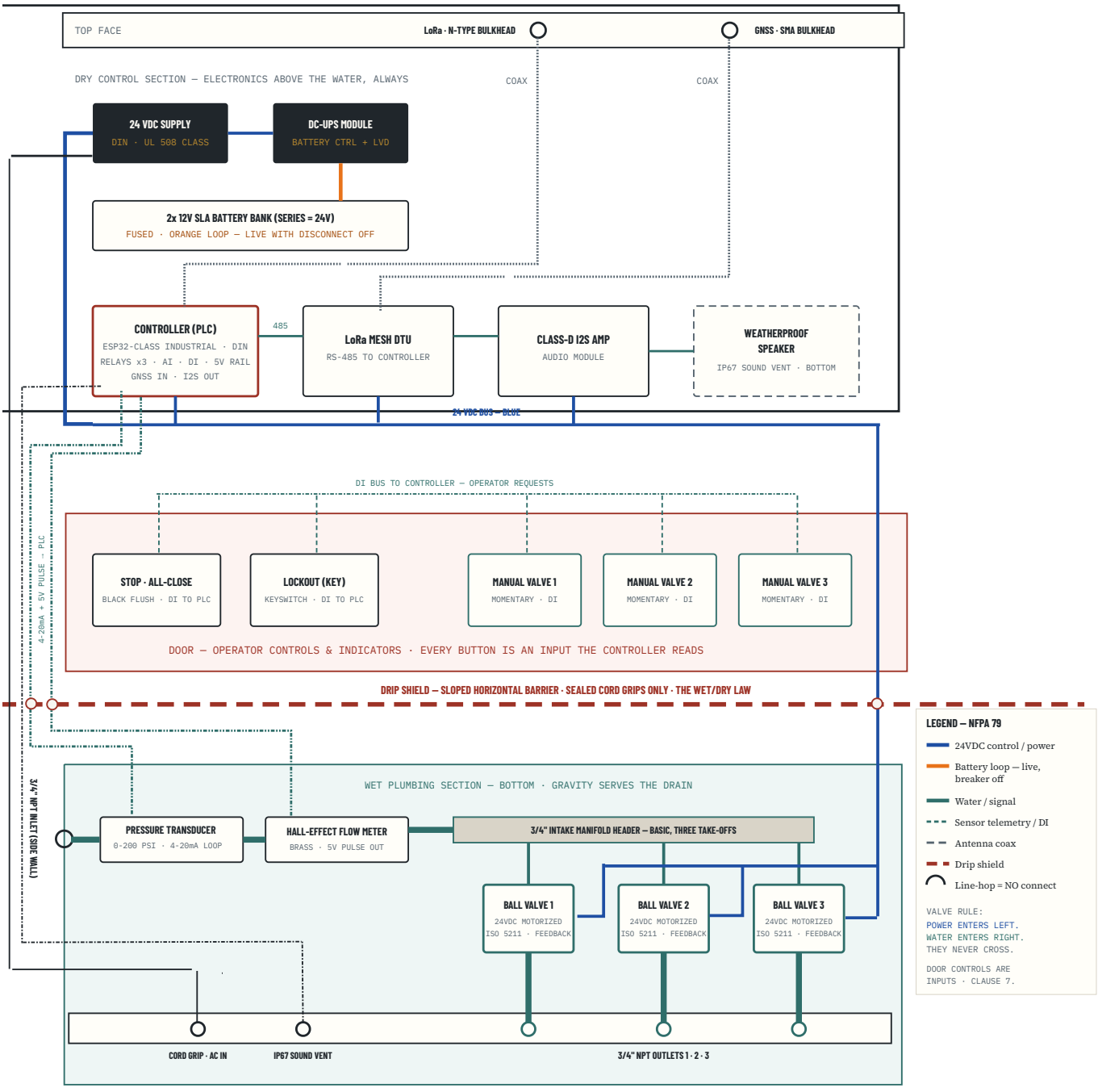


POWER LOSS DOES NOT APPEAR IN THIS FLOW

AUTO-RETURN VALVES CLOSE BY PHYSICS. THERE IS NO SOFTWARE STEP, AND SOFTWARE MUST NOT ASSUME IT CAN PREVENT OR DELAY IT.

EVERY DECISION THAT FAILS HAS A DEFINED DESTINATION, AND EVERY DESTINATION RETURNS. THE FLOW HAS NO TERMINAL FAILURE.

THIS SHEET IS THE ENCLOSURE. WHAT IS HIGH ON THE PAGE IS HIGH IN THE BOX. THE RED DASHED LINE IS REAL. POWER RUNS LEFT OF EACH VALVE, WATER RUNS RIGHT — THEY NEVER CROSS.



INSTRUCTIONAL — BUILD TO THE HARDWARE ANNEX · NAMED PARTS LIVE IN THE ANNEX PARTS CATALOGUE · THE REGISTER GOVERNS

FREE LIKE A FIRE CODE

THE STRIP IS THE CONTRACT: PANEL WIRING LANDS ON TOP IN NFPA COLORS, FIELD WIRING LANDS BELOW IN FACTORY COLORS.

PANEL SIDE (TOP) — BLUE CONTROL · WHT/BLU COMMON

TB1	TB2	TB3	TB4	TB5	TB6	TB7	TB8	TB9	TB10	TB11	TB12	TB13	TB14	TB15	TB16	TB17	TB18	TB19	TB20	G1	G2
M+	M−	LCOM	OPN	CLS	M+	M−	LCOM	OPN	CLS	M+	M−	LCOM	OPN	CLS	XD+	XSIG	5V+	PLS	FCOM	0V	PE
VALVE 1 TRACK					VALVE 2 TRACK					VALVE 3 TRACK					XDCR			FLOW METER		GROUNDS	

FIELD SIDE (BOTTOM) — FACTORY PIGTAIL COLORS THROUGH IP68 GRIPS

LANDING SCHEDULE — VALVE 1 TRACK (REPEAT PATTERN FOR TRACKS 2 AND 3)

TB1 M+ · field: valve RED (drive +) · panel: BLUE from PLC RELAY 1 OUT — the only drive path

TB2 M− · field: valve BLACK (drive −) · panel: WHT/BLU to 0V common (G1)

TB3 LCOM · field: valve GRN/WHT (limit common) · panel: BLUE from 24V+ bus — wets the switches

TB4 OPN · field: valve YELLOW (open limit) · panel: BLUE to PLC IN_0, teed to ACT lamp group (see box)

TB5 CLS · field: valve BLUE (closed limit) · panel: BLUE to PLC IN_1, teed to RDY lamp chain (see box)

TRACK 2 → IN_2/IN_3 · TRACK 3 → IN_4/IN_5 · SAME PATTERN, NEXT CHANNELS

SENSOR BLOCKS

TB16 XD+ · transducer RED · panel: BLUE from 24V+ bus (4-20mA loop power)

TB17 XSIG · transducer BLACK/shield · panel: BLUE to PLC AI_0 — the loop return IS the signal

TB18 5V+ · panel: from PLC 5V logic rail — noise-isolated feed for the meter

TB19 PLS · flow meter pulse out · panel: BLUE to PLC IN_6 — the totalizer counts here

TB20 FCOM · flow meter common · panel: WHT/BLU to 0V (G1)

LAMP LOGIC — WIRED TRUTH, THREE VALVES

RDY (white): all three CLOSED switches in SERIES → lamp lights only when EVERY valve confirms closed.

ACT (red): all three OPEN switches in PARALLEL → lamp lights when ANY valve is physically open.

RELAY LOGIC IN COPPER — THE LAMPS DESCRIBE THE SYSTEM, NOT ONE VALVE, AND NEVER THE SOFTWARE.

GROUND BLOCKS — TWO GROUNDS, NEVER MIXED

G1 0V DC MASTER (WHT/BLU block) — supply negative, DC-UPS negative, valve and sensor returns tie here

G2 CHASSIS EARTH (GREEN rail-bond block) — AC mains ground, enclosure shell, door braid. Bonds to the DIN rail itself.

FIG. 7 — REFERENCE INSTALLATION, CONCEPT VIEW

ILLUSTRATIVE RENDERING OF AN INSTALLED RESIDENTIAL-CLASS NODE UNDER EMBER CAST



CONCEPT ART — NOT A CONSTRUCTION DRAWING

This view illustrates a residential-class node as installed: the sealed wall-mounted enclosure with its external supply connection, the distribution line routed to the roof, and ridge-mounted heads wetting the roof plane and the structure's immediate surroundings during an ember cast. Dimensions, component placement, finish, head count, and head spacing are illustrative. The governing technical disclosure is FIG. 5, the node master schematic, and the Hardware Annex.